SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	K. Yamuna	**Reg. No.:**	192511013
Date:	09/10/2026	**Duration:**	60 minutes

Code of Conduct

*I **k Yamuna(192511013)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Yamuna K

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

ANNEXURE A – AGILE SPRINT PLANNING AND USER STORY WORKSHOP

PROJECT TITLE: SMART PARKING MANAGEMENT SYSTEM (TRANSPORT)

1. IDENTIFY THE PROJECT VISION AND STAKEHOLDERS

1.1 Project Vision

The Smart Parking Management System is designed to help drivers find available parking spaces in malls and college campuses. It provides real-time parking slot availability, advance booking, automated entry and exit billing, and occupancy reports. The system reduces the time spent searching for parking and helps parking operators manage parking spaces efficiently.

1.2 Project Objectives

1. To display available and occupied parking slots in real time.
2. To allow drivers to reserve parking spaces in advance.
3. To automate vehicle entry, exit, and parking fee calculation.
4. To reduce traffic congestion and waiting time.
5. To generate parking occupancy and revenue reports.

1.3 Key Stakeholders and Their Roles

1. Drivers / Customers: Check parking availability, reserve slots, and pay parking fees.

2. Parking Operators: Monitor parking spaces, manage bookings, and oversee parking operations.

3. Parking Attendants: Verify vehicles and assist with entry and exit procedures.

4. System Administrator: Manage user accounts, permissions, and system security.

5. Mall / Campus Management: Monitor parking usage, revenue, and overall system performance.

6. Payment Gateway Provider: Process digital parking payments securely.

7. Maintenance Team: Maintain parking sensors, cameras, automated gates, and other equipment.

2. CREATE EPICS AND USER STORIES

An epic is a large functionality of the system, while a user story describes a specific feature from the user's perspective.

Epic 1: User Registration and Login

User Story 1:

As a driver, I want to register and log in to the system, so that I can access parking services securely.

User Story 2:

As an administrator, I want to manage user accounts, so that only authorized users can access the system.

Epic 2: Real-Time Parking Slot Availability

User Story 3:

As a driver, I want to view available parking slots in real time, so that I can find parking quickly.

User Story 4:

As a parking operator, I want to view occupied and available slots, so that I can manage parking spaces efficiently.

Epic 3: Advance Parking Booking

User Story 5:

As a driver, I want to book a parking slot in advance, so that I can reserve a space before arriving.

User Story 6:

As a driver, I want to cancel my booking, so that I can release the slot when my plans change.

Epic 4: Automated Entry, Exit, and Billing

User Story 7:

As a driver, I want the system to record vehicle entry and exit times, so that my parking duration can be calculated accurately.

User Story 8:

As a driver, I want the parking fee to be calculated automatically, so that I can pay the correct amount.

Epic 5: Parking Analytics and Reports

User Story 9:

As a parking operator, I want to view occupancy reports, so that I can understand parking usage.

User Story 10:

As mall or campus management, I want to view parking revenue reports, so that I can make better management decisions.

3. DEFINE ACCEPTANCE CRITERIA

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete.

User Story 1: User Registration and Login

- Given a registered user, when valid login details are entered, then the system should allow access.
- Given invalid login details, when the user attempts to log in, then the system should display an error message.

User Story 2: User Account Management

- Given an administrator is logged in, when the administrator updates an account, then the system should save the changes.
- Only authorized administrators should be allowed to manage user accounts.

User Story 3: Real-Time Slot Availability

- Given a driver opens the parking dashboard, when the system retrieves the latest parking data, then available and occupied slots should be displayed.
- When a slot changes its status, the system should update the displayed status within 10 seconds under normal operating conditions.

User Story 4: Operator Parking Dashboard

- Given a parking operator is logged in, when the operator opens the dashboard, then the total, available, and occupied slots should be displayed.

User Story 5: Advance Parking Booking

- Given a parking slot is available, when a driver selects a valid date and time and confirms the booking, then the system should reserve the slot and generate a confirmation.
- The system should prevent two drivers from booking the same slot for overlapping time periods.

User Story 6: Booking Cancellation

- Given a driver has an eligible booking, when the driver cancels it, then the system should update the booking status and release the slot according to the cancellation policy.

User Story 7: Vehicle Entry and Exit

- Given a vehicle has valid parking authorization, when it enters or exits, then the system should record the corresponding time.
- Entry and exit timestamps should be stored for parking duration calculation.

User Story 8: Automatic Billing

- Given a vehicle completes its parking session, when the exit time is recorded, then the system should calculate the parking fee according to the configured rates.
- The final fee should be displayed before payment.

User Story 9: Occupancy Reports

- Given a parking operator is logged in, when an occupancy report is requested, then the system should display parking usage for the selected period.

User Story 10: Revenue Reports

- Given management selects a date range, when a revenue report is requested, then the system should display the recorded parking revenue for that period.

4. PRIORITIZE THE PRODUCT BACKLOG

The product backlog contains all the features planned for development. The features are prioritized according to business value, customer needs, and dependencies.

Priority 1 – High

1. User registration and login.
2. Real-time parking slot availability.
3. Advance parking booking.
4. Vehicle entry and exit recording.
5. Automatic parking fee calculation.

Priority 2 – Medium

6. Parking operator dashboard.
7. Booking cancellation.
8. Parking occupancy reports.

Priority 3 – Low

9. Parking revenue analytics.
10. Predictive parking demand analysis.

Justification for Prioritization

High-priority features are essential for the basic operation of the system. User authentication provides secure access, real-time availability helps drivers find parking, and advance booking allows them to reserve spaces. Entry, exit, and billing features support the complete parking process. Advanced analytics can be implemented after the core features are completed.

5. PLAN THE SPRINT BACKLOG

A sprint backlog is a collection of high-priority user stories and development tasks selected for a sprint.

Sprint Duration: 2 Weeks

Sprint Number: Sprint 1

5.1 Sprint Goal

To develop the basic Smart Parking Management System with secure user login, real-time parking slot availability, and advance parking booking.

5.2 Selected User Stories

US01 – User Registration and Login

Priority: High

US03 – Real-Time Parking Slot Availability

Priority: High

US05 – Advance Parking Booking

Priority: High

5.3 Development Tasks

US01 – User Registration and Login

1. Design the registration and login pages.
2. Validate user input.
3. Implement user authentication.
4. Test login and registration functionality.

US03 – Real-Time Parking Slot Availability

1. Design the parking dashboard.
2. Create the parking slot database.
3. Retrieve available and occupied slot information.
4. Display the latest slot status on the dashboard.
5. Test parking status updates.

US05 – Advance Parking Booking

1. Design the parking booking page.
2. Implement date and time selection.
3. Check slot availability before confirming a booking.
4. Prevent duplicate bookings.
5. Generate booking confirmation.
6. Test the booking functionality.

5.4 Sprint Schedule

Days 1–2: Requirement analysis, sprint planning, and interface design.

Days 3–5: Development of registration and login functionality.

Days 6–8: Development of real-time slot availability and booking functionality.

Days 9–10: Integration, testing, error correction, and sprint review.

6. ESTIMATE EFFORT USING STORY POINTS

Story points are used to estimate the relative effort, complexity, and uncertainty involved in completing a user story. The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for estimation. Planning Poker can be used by team members to discuss and agree on the story points.

6.1 Story Point Estimation

US01 – User Registration and Login

Estimated Story Points: 3

Justification: This feature requires interface development, input validation, and authentication. Its complexity is relatively low.

US03 – Real-Time Parking Slot Availability

Estimated Story Points: 5

Justification: This feature requires database integration, parking status retrieval, and timely dashboard updates.

US05 – Advance Parking Booking

Estimated Story Points: 8

Justification: This feature requires booking logic, availability verification, date and time selection, and prevention of duplicate reservations.

Total Estimated Story Points: 16

6.2 Estimation Technique

The team uses Planning Poker to estimate each user story. Each member suggests a story point value based on the effort and complexity involved. The team discusses different estimates and agrees on a final value.

The estimates are preliminary and can be adjusted based on team capacity, technical difficulties, and testing results.

7. PRESENT THE SPRINT GOAL AND EXPECTED DELIVERABLES

7.1 Sprint Goal

The goal of Sprint 1 is to develop a working basic version of the Smart Parking Management System that allows users to log in, view available parking slots, and book a parking space in advance.

7.2 Expected Sprint Deliverables

1. A working user registration and login module.
2. A parking dashboard displaying available and occupied slots.
3. An advance booking feature with booking confirmation.
4. A database for storing user, parking slot, and booking details.
5. Tested and integrated features with basic error handling.

7.3 Definition of Done

A user story is considered complete when:

1. All acceptance criteria are satisfied.
2. The feature is integrated into the application.
3. Functional testing is completed successfully.
4. Critical defects are resolved.

5. The code and relevant documentation are reviewed.

CONCLUSION

The Smart Parking Management System provides an efficient solution for finding and reserving parking spaces in malls and college campuses. Agile sprint planning helps the development team organize requirements, prioritize features, estimate effort, and deliver working software in stages. The first sprint establishes the foundation for user login, real-time parking availability, and advance booking. Future sprints can introduce automated entry and exit, billing, occupancy analytics, and advanced parking management features.